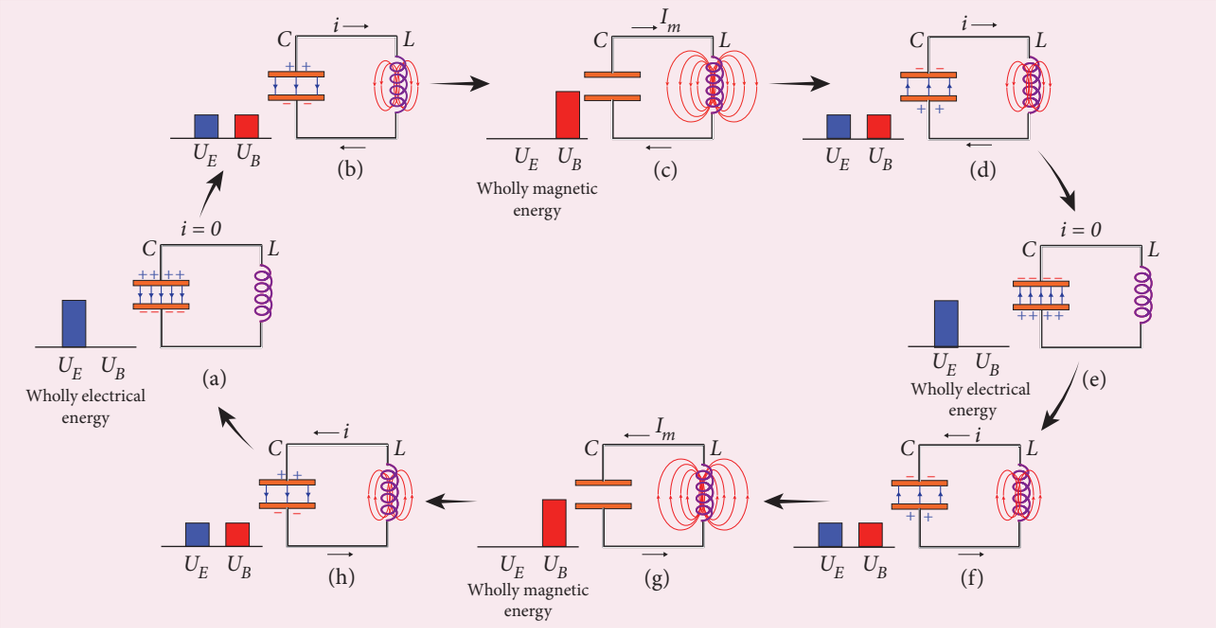


Figure 4.52 LC oscillations

maximum current flowing in the circuit. The total energy is wholly magnetic (Figure 4.52(c)).

Even though the charge in the capacitor is zero, the current will continue to flow in the same direction because the inductor will not allow it to stop immediately. The current is made to flow with decreasing magnitude by the collapsing magnetic field of the inductor. As a result of this, the capacitor begins to charge in the opposite direction. A part of the energy is transferred from the inductor back to the capacitor. The total energy is the sum of the electrical and magnetic energies (Figure 4.52(d)).

When the current in the circuit reduces to zero, the capacitor becomes fully charged in the opposite direction. The energy stored in the capacitor becomes maximum. Since the current is zero, the energy stored in the inductor is zero. The total energy is wholly electrical (Figure 4.52(e)).

The state of the circuit is similar to the initial state but the difference is that the capacitor is charged in opposite direction. The capacitor then starts to discharge through the inductor with anti-clockwise current. The total energy is the sum of the electrical and magnetic energies (Figure 4.52(f)).

As already explained, the processes are repeated in opposite direction (Figure 4.52(g) and (h)). Finally, the circuit returns to the initial state (Figure 4.52(a)). Thus, when the circuit goes through these stages, an alternating current flows in the circuit. As this process is repeated again and again, the electrical oscillations of definite frequency are generated. These are known as *LC* oscillations.

In the ideal *LC* circuit, there is no loss of energy. Therefore, the oscillations will continue indefinitely. Such oscillations are called undamped oscillations.

12th Physics_Vol-1_Unit 4_English Medium.indd 252 23-12-2022 17:59:01